

# System Architecture - N100 Financial Intelligence

The system follows a modular 3-tier architecture:

1. Data Layer (ETL): SQLite database populated from raw Excel files.
2. Business Logic Layer (Analytics & NLP): KMeans clustering, heuristic rule engine, cashflow KPIs, and dynamic screener.
3. Presentation Layer (API & Dashboard): FastAPI provides 16 REST endpoints. Streamlit provides an interactive 8-screen dashboard.
4. Reporting Engine: ReportLab and Matplotlib generate automated batch PDFs.